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Quantifying cultural ecosystem services in urban parks using social media and large language models: Insights for Beijing's Garden City initiative

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ABSTRACT

Urban parks are vital components of sustainable cities, providing cultural ecosystem services (CES) that enhance residents' well-being. Yet understanding how people perceive these intangible benefits remains challenging. This study refines existing CES research by integrating large-scale social media analytics with large language models (LLMs) to capture and interpret public perceptions of parks in Beijing's core districts. Drawing on more than 46,000 user-generated comments, the LLM-based framework identifies six CES categories (i.e., historical & cultural, aesthetic, recreational, social interaction, educational, and spiritual values) and evaluates their sentiment tendencies. The results reveal that aesthetic value is the most positively perceived dimension, while recreational, educational, and social interaction values show moderate emotional intensity, indicating potential areas for experiential enhancement. Historical & cultural and spiritual values, though less frequently mentioned, generate strong positive affect when recognised. Integrating these findings through Importance-Performance Analysis highlights clear renewal priorities for Beijing's Garden City initiative. Methodologically, the study demonstrates how LLMs can classify Chinese-language social media text with high accuracy ($F1 = 0.8945$) and minimal manual annotation, offering a scalable, interpretable, and cost-effective approach for CES assessment. The combined use of social media data, semantic modelling, and performance analysis provides transferable evidence for guiding park management and renewal in other high-density urban contexts.

1. Introduction

Urban sustainability is closely intertwined with environmental quality, and urban parks and green spaces play a pivotal role in achieving it (Arnberger, 2012; La Rosa et al., 2017). Urban parks are among the most accessible and valued forms of urban green space. Beyond their ecological functions, they provide opportunities for recreation, aesthetic enjoyment, social interaction, and cultural continuity, thereby playing a central role in sustaining urban residents' well-being (Coldwell and Evans, 2018; Hand et al., 2017; Yang et al., 2021;

Zhang and Zhou, 2018). These non-material benefits are often difficult to quantify, yet they are essential for guiding park planning and renewal in rapidly evolving urban environments.

The framework of cultural ecosystem services (CES)¹ offers a systematic approach to capture these non-material benefits (De Groot et al., 2002; Millennium Ecosystem Assessment, 2005). CES encompass aesthetic, recreational, spiritual, educational, and cultural values that emerge from human interactions with ecosystems (Costanza et al., 1997; Millennium Ecosystem Assessment, 2005). Over the past decade, CES in urban contexts (including parks) have been widely examined. Rather

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¹ In parallel to CES, related frameworks such as the Social Values for Ecosystem Services (SolVES) model have been widely applied to map and quantify perceived values of green spaces (Duan and Xu, 2022; Sherrouse et al., 2014; Sun et al., 2019). In this study, we adopt CES as the overarching framework because it allows us to capture multiple non-material benefits and link them directly to social media narratives. This approach avoids conflation with SolVES while building on its conceptual lineage.

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than claiming a lack of research, this study builds upon this extensive scholarship by addressing several persistent challenges. First, most existing research relies on surveys or manually coded social media analyses, which are difficult to scale up and often overlook contextual nuance in everyday language. Second, how CES perceptions vary across different types of parks² and spatial contexts remains underexplored, particularly in cities where heritage, density, and public space renewal intersect. Third, there is limited integration between CES analytics and practical planning frameworks that can inform renewal and management strategies. Addressing these gaps requires methodological innovation capable of capturing fine-grained public perceptions while linking them to differentiated park-management priorities.

In recent years, urban greening strategies have increasingly shifted from a primary focus on ecological provision towards enhancing residents' lived experience, cultural identity, and everyday well-being. Beijing offers a particularly relevant context for examining this shift through its Garden City Initiative (2023–2035), which explicitly seeks to integrate ecological quality with cultural, recreational, and social functions across the capital's park system (Xu et al., 2012). However, despite the centrality of these human-centred objectives, empirical evidence on how residents perceive and value the CES provided by Beijing's parks remains limited. Without such evidence, renewal strategies risk privileging physical upgrading over experiential quality, especially in the historic core of the city, where landscape heritage, cultural meaning, and everyday leisure needs are deeply intertwined.

Conventional methods for assessing CES, such as questionnaires or participatory mapping, have generated valuable insights but are often resource-intensive, slow to update, and limited in spatial and demographic coverage. To address these constraints, a growing body of research has turned to social media data as a scalable and dynamic source for capturing public perceptions of urban parks across space and time (Zhang and Zhou, 2018; Wood et al., 2013). Studies have demonstrated the feasibility of such approaches: Plunz et al. (2019) identified differences in emotional expression inside and outside parks in New York using Twitter data, while Kong et al. (2022) showed how positive emotions varied across park types in Beijing based on Weibo posts. Other work has used review platforms such as TripAdvisor to classify CES categories through crowdsourced lexicons and keyword-based text analysis across large park systems (Kong et al., 2023), or combined social media text and image analysis to explore relationships between perceived CES, landscape features, and visitor characteristics (Chen et al., 2024).

Despite this progress, most existing approaches rely on keyword matching, rule-based lexicons, or pre-trained sentiment models, with CES categories often defined or weighted through expert-based scoring. While effective for broad pattern detection, these methods struggle to capture contextual nuance in everyday language and to distinguish overlapping or co-occurring CES dimensions. As a result, interpretability and category granularity remain constrained, limiting the usefulness of CES analytics for informing differentiated planning and management decisions. Addressing these limitations requires methodological frameworks that can capture context-sensitive meanings at scale while remaining transparent and reproducible.

Recent advances in large language models (LLMs) provide a promising way to address these challenges. By modelling language at a semantic and contextual level, LLMs are capable of interpreting nuanced expressions, recognising co-occurring meanings, and supporting multi-label classification within a single text (Wu et al., 2024). Through

explicit prompt design, their classification logic can be made transparent and conceptually aligned with planning frameworks, improving interpretability and practical relevance. In addition, through prompt engineering and few-shot learning, LLMs can perform context-sensitive text classification without costly retraining or extensive manual annotation (Sahoo et al., 2025). Early applications in planning have already shown promise: for example, Zhong et al. (2024) used LLMs to parse multimodal social media data to explore visitors' emotional tendencies and activity richness in urban parks, while Luo et al. (2025) demonstrated the potential of prompt-based and few-shot approaches for CES classification. Building on these developments, this study employs LLMs as the primary tool for classifying perceptions of CES in urban parks, enabling fine-grained semantic interpretation while substantially reducing reliance on manual coding.

This study applied these advances to examine CES in 24 parks across Beijing's core districts. Drawing on multi-source social media data, it integrates LLM-based text classification, sentiment analysis, and Importance–Performance Analysis (IPA) to evaluate how residents perceive and emotionally evaluate the cultural services provided by parks. Three research questions guide the analysis: (1) How can CES in urban parks be systematically identified and classified using LLMs? (2) How do residents perceive and emotionally evaluate different CES dimensions across park types? (3) What management and planning recommendations can inform park renewal under the Garden City initiative?

Notably, this study makes three contributions. First, it provides one of the earliest large-scale assessments of CES in urban parks using social media data, encompassing over 46,000 user-generated comments. Second, it demonstrates the integration of LLM-based text classification, sentiment analysis, and IPA as an efficient and reproducible workflow linking public-perception analytics with park-management priorities. Third, by linking empirical findings to planning practice, it highlights that aesthetic value is the most positively perceived dimension, while recreational, educational, and social interaction values warrant targeted enhancement. These advances refine methodological practice in CES evaluation and provide evidence-based insights for urban-park renewal in Beijing and other high-density cities worldwide.

2. Materials and methods

2.1. Study area

This study selected a total of 24 urban parks within the Core Area of Beijing (CAB) as the empirical focus (Fig. 1). CAB includes a total of 32 neighbourhood-level administrative units and a total area of 92.5 square kilometres. Selection was guided by three criteria: (1) Park typology: ensuring representation of Historical, Comprehensive, Community, and Specialised parks, consistent with Beijing's official park classification³; (2) Scale and function: including both large, city-level parks of international and national significance (e.g., Beihai Park, Temple of Heaven Park) and smaller community-scale parks that serve local residents' daily needs; and (3) Public prominence: parks with high levels of recognition, accessibility, and administrative importance. Collectively, these 24 parks account for the most visited and symbolically significant green spaces in Beijing's historic core, while also covering the main functional categories of the city's wider park system. Although they do not capture every park type citywide, they provide a representative

² Based on prior research, we anticipated differences across park types. In particular, we hypothesised that historical parks would yield higher levels of perceived historical-cultural value, whereas comprehensive or community parks would emphasise recreational and social interaction values. Testing these expectations provides an opportunity to demonstrate the utility of CES analysis for distinguishing functional profiles across heterogeneous parks.

³ Park typologies in this study follow the official *Beijing Park List* (2024) issued by the municipal authority. In this administrative classification, certain parks (e.g., Beijing Zoo) are designated as historical parks based on their historical origin, heritage attributes, and governance status, even though their contemporary functions may emphasise recreation or education. This study adopts the official classification to ensure consistency with municipal planning and management frameworks.

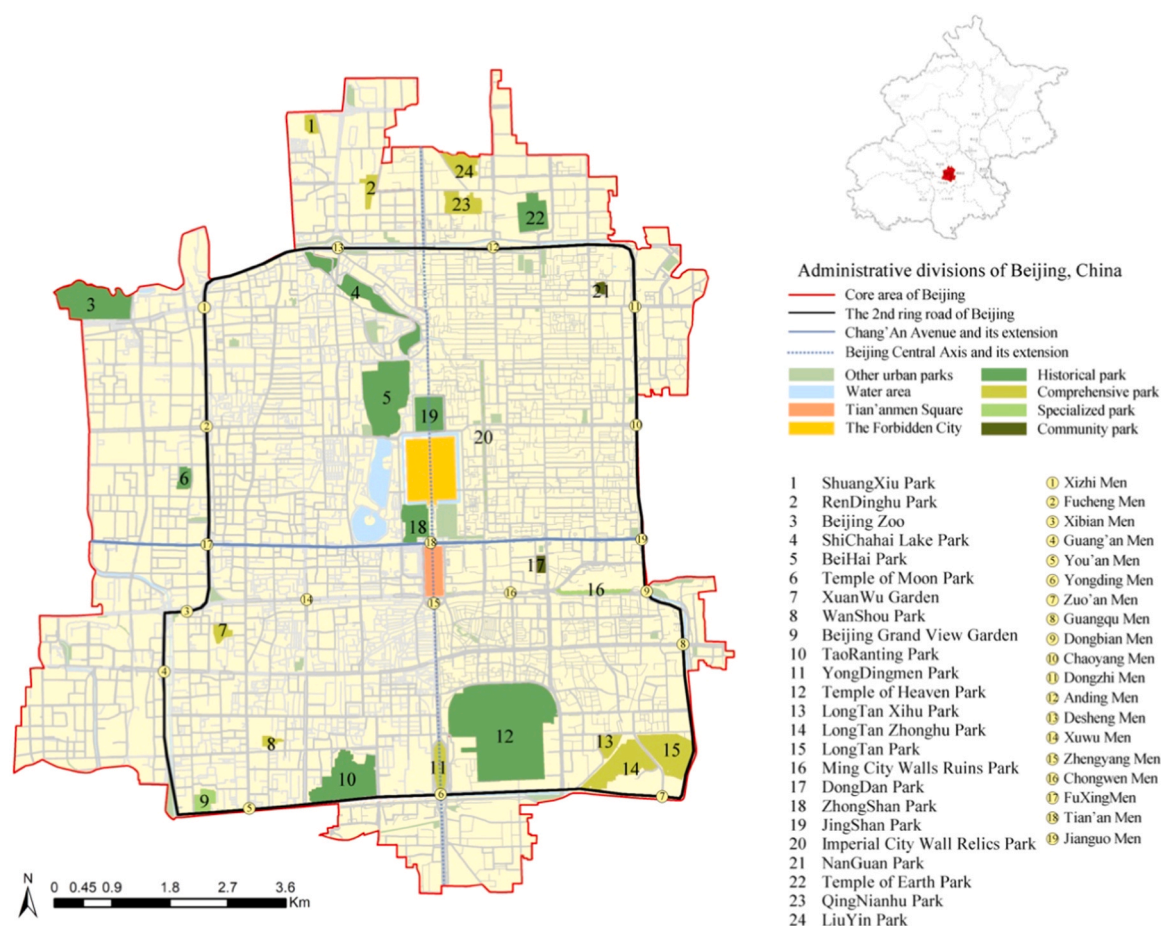


Fig. 1. Location and classification of the case study parks in the Core Area of Beijing (CAB).

cross-section of Beijing's park landscape, spanning heritage landscapes, multifunctional urban parks, and neighbourhood-level spaces. This breadth ensures that the findings offer meaningful insights into CES across Beijing's diverse urban park system. Further details on park typology, scale, and distribution are available in [Supplementary Material A](#).

2.2. Data and methods

The overall research framework is illustrated in [Fig. 2](#). The analysis followed a stepwise process. First, user comments on the 24 parks were collected from Sina Weibo and Xiaohongshu (RedNote) and processed through lexical cleaning and segmentation. Second, an index system of CES was established by integrating insights from high-frequency word analysis with established CES typologies. Third, the comments were classified into CES categories using the Moonshot-v1-8k large language model, and the perceived frequency of each CES type was calculated. Fourth, sentiment tendencies were assessed using the Baidu AI Cloud Sentiment Analysis API. Finally, statistical analyses (i.e., correlation and Importance-Performance Analysis (IPA)) were conducted to derive evidence-based recommendations for urban-park renewal and management, with implications for resident well-being.

2.2.1. Step 1. Social media data collection

User-generated content was obtained from Sina Weibo and Xiaohongshu (RedNote), two of China's largest social media platforms that host diverse demographic groups and are widely used in urban-perception research. Data collection covered the period from May 2023 to May 2024. To identify relevant posts, search queries were constructed using each park's official Chinese name. The complete list of

search keywords is provided in [Supplementary Material A](#). Data were retrieved via Python-based web scraping and limited to publicly available posts.⁴

To ensure quality, duplicate records, advertisements, and promotional content were removed through rule-based filters. Posts containing only URLs, emojis, or irrelevant topics were excluded. A random sample of 597 entries was manually checked, confirming that most referred directly to visitors' park experiences. After cleaning, the final dataset comprised 46,243 valid comments associated with the selected parks.

2.2.2. Step 2. High-frequency word analysis and CES classification

Step 2a. High-frequency word analysis. To identify the semantic features of user perceptions, we performed a high-frequency word analysis on the cleaned dataset using *Jieba* for Chinese word segmentation. After removal of stopwords (e.g., conjunctions, pronouns, irrelevant words), we extracted a total of 695 distinct high-frequency words, each appearing more than 100 times across the dataset.

Step 2b. CES classification. These words provided the empirical basis for linking user-generated content to categories of CES. Following established frameworks ([Millennium Ecosystem Assessment, 2005](#)) and previous studies, six CES categories were identified: historical & cultural (H&C) value, aesthetic value, recreational value, social interaction value, educational value, and spiritual value. Representative keywords

⁴ All data were drawn from publicly accessible posts. Usernames were anonymised during processing, and no private or sensitive content was collected. Data collection complied with the platforms' terms of service and was conducted solely for academic research purposes; therefore, no separate ethical approval was required ([Li et al., 2020; Zhang et al., 2024](#)).

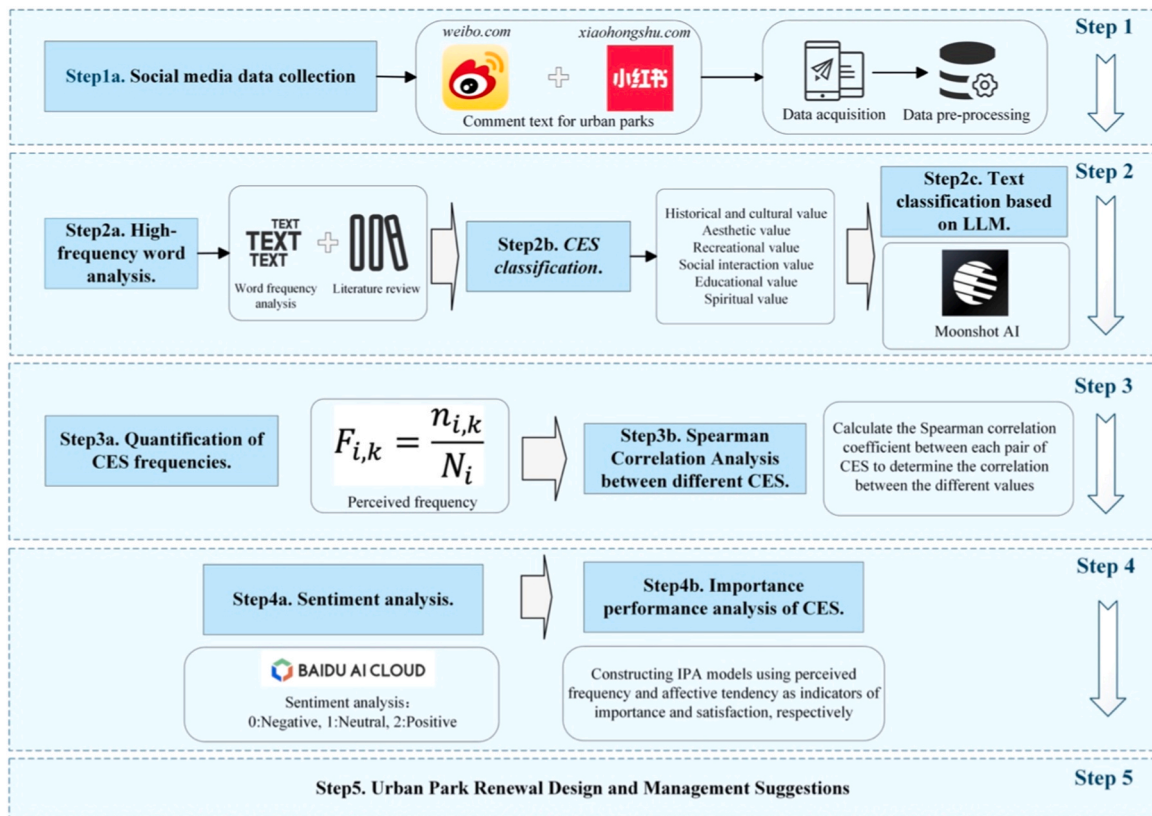


Fig. 2. Research framework.

associated with each CES category are summarised in Table 1, and a more extensive keyword mapping is provided in Supplementary Material B.

Step 2c. Text classification based on LLM. The classification of comments into CES categories was conducted using LLMs. Compared with traditional machine learning methods, which require extensive manually labelled training data (Xu and Goodacre, 2018), LLMs can perform text categorisation through prompt engineering and few-shot learning, thus reducing the workload of manual annotation.

2.2.2.1. Model selection. Given the Chinese-language context, it was important to select a model pre-trained on large Chinese corpora to minimise language bias. Recent comparative work (see, e.g., Luo et al., 2025) demonstrated that Moonshot-v1-8k (developed by Moonshot AI, <https://www.moonshot.cn/>) achieved the strongest performance for CES perception tasks in Chinese. Accordingly, we adopted Moonshot-v1-8k for text classification. The model was accessed via its official API and the OpenAI-compatible Python library. Supplementary Material C provides further technical details of the model architecture.

2.2.2.2. Prompt engineering. Prompt engineering was applied according to best-practice few-shot learning principles (Luo et al., 2025). Each prompt contained: (1) a clear task definition instructing the model to classify comments into CES types; (2) concise descriptions of the six CES categories; (3) at least one exemplar comment per category to guide classification (see Table 2); and (4) a role specification designating the model as an “expert in urban and rural planning”. The Moonshot-v1 model was configured with a temperature parameter of 0.3, following the official usage documentation, to ensure balanced accuracy and diversity in outputs. The model generated structured JSON responses for each comment, enabling reliable parsing and subsequent statistical analysis. No additional model training was performed; classification relied entirely on in-context learning capabilities (Fig. 3).

2.2.2.3. Validation. A stratified random sample of 597 comments⁵ were independently annotated by two coders, proportionally distributed across all 24 parks to ensure representativeness. Agreement was very high (Cohen’s $\kappa = 0.935$) using SPSS 27.0. To improve transparency, accuracy, precision, recall, and F1-score were all computed. Detailed results and confusion matrices are provided in Supplementary Material C. Mean F1 = 0.8945 indicates robust overall performance (Table 3), with aesthetic and recreational values > 0.95 and educational value lower (0.7556) because of conservative recall.⁶

2.2.3. Step 3. Quantification of CES frequencies and inter-value correlations

Step 3a. Quantification of CES frequencies. Following classification, the perceived frequency of each CES value was calculated for every park. The CES of urban parks is reflected in the public’s sensitivity, concern, and perception of various values. The higher the perceived frequency, the higher the sensitivity, attention, and perception. Therefore, we can have a deeper understanding of the CES of urban parks by quantifying the perceived frequency. The frequency of a given CES type in a park was computed as:

$$F_{i,k} = \frac{n_{i,k}}{N_i}$$

where $n_{i,k}$ is the number of comments for CES type k in park i , and N_i is the total number of comments for park i .

Step 3b. Spearman correlation analysis between different CES. Due to the

⁵ Although this validation set represents a relatively small proportion of the total dataset, its stratified sampling design and the very high inter-annotator agreement confirm that it is sufficient for assessing model reliability.

⁶ A conservative bias that under-detects some relevant educational comments while maintaining high precision.

Table 1
Classification of cultural ecosystem services (CES).

No.	CES type	CES definition	References for indicator settings	Representative keywords
1	Historical & cultural (H&C) value	Inheriting ancestral wisdom and knowledge, traditions, and cultures with significant natural and human historical value.	Chen et al. (2024); Guo et al. (2023); Sun et al. (2019); Wang et al. (2021)	Culture, ruins, the Hall of Prayer for Good Harvests, etc
2	Aesthetic value	Unique sights, sounds, and smells, including natural and human-made beauty.	Chen et al. (2024); Dai et al. (2019); Guo et al. (2023); Wang et al. (2021)	Photos, chrysanthemums, landscapes, etc
3	Recreational value	Places where people engage in recreational activities that generate leisure recreational behaviours and access to recreational services.	Chen et al. (2024); Dai et al. (2019); Sun et al. (2019); Wang et al. (2021)	Travel, activities, entertainment, etc
4	Social interaction value	Provides an environment for human interaction and promote friendly interaction among people.	Chen et al. (2024); Wang et al. (2021)	Friends, grandpa, children, etc
5	Educational value	The carrier of nature education and science popularization activities, enable people to acquire knowledge.	Chen et al. (2024); Dai et al. (2019); Guo et al. (2023); Wang et al. (2021)	Museums, book markets, teachers, etc
6	Spiritual value	To cultivate the sentiment, purify the mind, and obtain spiritual support.	Chen et al. (2024); Dai et al. (2019); Sherrouse et al. (2014)	Atmosphere, life, mood, etc

perceived frequencies showing a non-normal distribution, spearman

Table 2
Representative few-shot examples used to guide LLM-based multi-label classification of cultural ecosystem services (CES).

Comment	Historical & cultural (H&C) value	Aesthetic value	Recreational value	Social interaction value	Educational value	Spiritual value
"The spring in Beijing is so beautiful. Come to Beihai Park, walk along the lakeshore, take photos, and rest on the benches. There are many painters sketching and writers creating. I really like this feeling – it's so cosy! I also took many beautiful photos and feel happy every day."	0	1	1	1	0	1
"Took the kids to visit the Temple of Heaven Park. The park is more attractive to children than the National Museum. #Beijing Temple of Heaven Park"	0	0	1	1	0	0
"The Ming Dynasty City Wall Site in Beijing, dating back over 580 years, is a component of the original inner city wall of Beijing. Standing beneath the city wall, one can feel the weight of history and the vicissitudes of time. Strolling through the park, surrounded by ancient trees, lush green grass, and a wide variety of flowers and trees, one is greeted by a splendid display of blossoms, birdsong, and fragrance in this spring season. It is a rare and wonderful leisure destination."	1	1	1	0	0	0
...	...	...	...	...	...	...

Notes: 1 = value present; 0 = value absent. At least one exemplar comment was provided for each CES category in the prompts, enabling the model to infer classification logic through few-shot learning. Full prompt content and JSON output structure are detailed in [Supplementary Material C](#).

correlation analyses were conducted for each type of CES using SPSS 27.0. To explore interrelationships among CES categories, Spearman correlation analysis was applied to the park-level frequency data. This enabled assessment of whether particular CES values tended to co-occur in visitor perceptions (e.g., aesthetic value and recreational value).

2.2.4. Step 4. Sentiment analysis and importance-performance analysis (IPA)

Step 4a. Sentiment analysis. Sentiment analysis was conducted to evaluate users' emotional responses to different CES categories. Text sentiment analysis involves the computational detection, analysis, and extraction of subjective expressions in user-generated text, including opinions, preferences, emotions, and related constructs. Each comment's polarity was computed using the Baidu AI Cloud Sentiment Analysis API, which applies deep neural networks trained on extensive Chinese-language corpora and achieves over 95 % accuracy in validated applications (Xie et al., 2021). This tool was selected for its proven robustness in handling short, colloquial social media texts common on Weibo and Xiaohongshu. Currently, many studies involving Chinese text sentiment analysis have been conducted using this platform (Chen et al., 2024; Li et al., 2024).

The model outputs four parameters: positive probability, negative probability, sentiment, and confidence, which collectively describe the probability distribution of emotional polarity. The categorical variable sentiment (0 = negative, 1 = neutral, 2 = positive) was used to compute the mean sentiment score for each CES type and park (see [Supplementary Material D](#)). All analyses were performed on cleaned Chinese text after removal of punctuation, emojis, and hashtags to minimise noise.

In addition, the significance of the differences test was used to determine whether there is a significant difference in the emotional disposition of urban parks. The hypothesis H0 of the significance of the difference test was that there was no significant difference in the emotions of visitors to different types of parks. In this study, the data were first tested to see if they were normally distributed in order to select the appropriate test, and it was found that the data were non-normally distributed, so it was not possible to use analysis of variance (ANOVA). Based on previous experience, non-parametric significance test (Kruskal-Wallis test) was used instead of ANOVA (Kong et al., 2022; Sia et al., 2022). The Kruskal-Wallis H-test was used to assess overall variation among park types, followed by Mann-Whitney U-tests for pairwise comparisons.

Step 4b. Importance-performance analysis of CES. Importance-Performance Analysis (IPA) evaluates the disparity between the perceived significance of a service and its actual performance (Martilla and James, 1977). To evaluate both importance and satisfaction

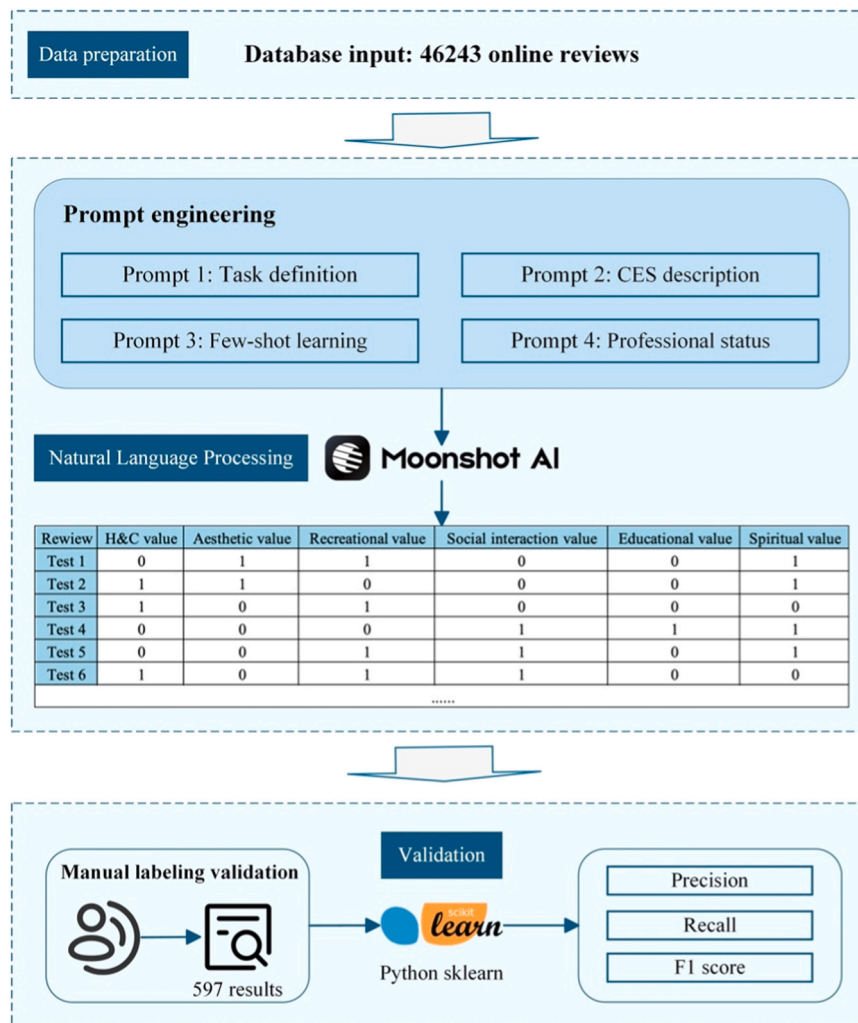


Fig. 3. Workflow for text analytics and evaluation.

Table 3
LLM performance in CES classification.

CES category	Accuracy	Precision	Recall	F1-Score
Historical & cultural value	0.9481	0.8421	0.9730	0.9028
Aesthetic value	0.9615	0.9451	0.9699	0.9573
Recreational value	0.9447	0.9507	0.9535	0.9521
Social interaction value	0.9615	0.8741	0.9615	0.9158
Educational value	0.9816	0.8947	0.6538	0.7556
Spiritual value	0.9397	0.8500	0.9189	0.8831
Overall	0.9562	0.8928	0.9051	0.8945

dimensions of CES, we employed Importance–Performance Analysis (IPA). “Importance” was proxied by the relative frequency of each CES value, while “Performance” was measured by the mean sentiment score of comments assigned to that CES. Each CES category was positioned on a two-dimensional IPA grid (importance $\times$ performance). The average values of importance and performance serve as the horizontal and vertical axes, respectively, with their intersection acting as the demarcation point, dividing the matrix into four zones. Distinct countermeasures are recommended based on the varying priorities of these quadrants (Das et al., 2022; Gai et al., 2022; Zhang et al., 2024), enabling identification of areas of strength, hidden strengths, low-priority values, and areas requiring improvement. IPA was implemented in SPSS 27.0.

Importantly, CES dimensions classified as “recommended improvement” within the IPA framework reflect relative under-performance in

relation to perceived importance within an overall positive sentiment context, rather than indications of dissatisfaction or negative public perception.

2.2.5. Step 5. Deriving urban park renewal and management suggestions

Finally, the IPA results were synthesised to derive practical recommendations for urban park renewal and management within the framework of Beijing’s Garden City initiative. Strategies were tailored according to CES quadrant positioning. High-performing values were maintained and refined, hidden strengths made more visible, low-priority values integrated with other functions, and recommended-improvement values prioritised for quality and inclusiveness enhancements. All recommendations align with municipal park-renewal guidelines, ensuring policy relevance and applicability.

3. Results

3.1. Quantitative characteristics of CES in the parks

By categorising the comment texts through the Moonshot AI model, this study derived both the number and perceived frequency of comments corresponding to each CES category across the 24 parks (Fig. 4).

All six CES categories were perceived in the various parks, but there are differences among different parks. Fig. 4a shows the total comment counts by park type: Historical parks accounted for 36,890 comments, Comprehensive parks for 7292, Specialised parks for 1546, and

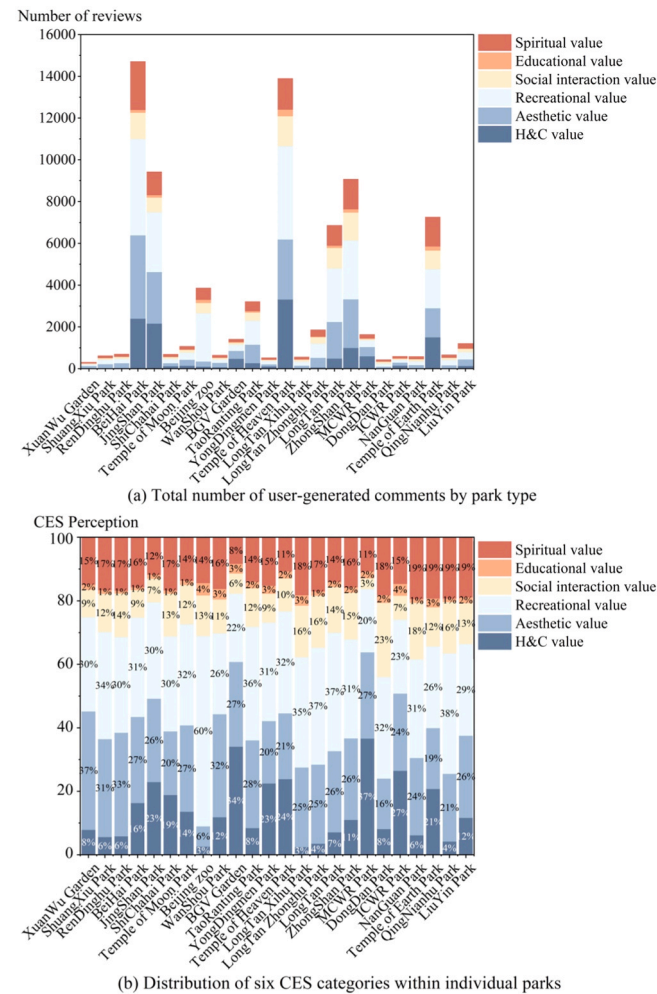


Fig. 4. Distribution of CES perceptions across park types and individual parks. (a) Total number of user-generated comments by park type; (b) Distribution of six CES categories within individual parks.

Community parks for 515. Fig. 4b shows the proportion of comments in each value category of each park to the total CES perception comments for individual parks.

As expected, Historical parks such as the Temple of Heaven Park (24 %) and Jingshan Park (23 %) yielded a high proportion of comments related to historical-cultural value. Similarly, two Specialised parks—the Imperial City Wall Relics Park (27 %) and the Ming City Wall Ruins Park (37 %)—showed strong alignment between their historic functions and perceived historical & cultural value.

In contrast, Beijing Zoo exhibited an unusual pattern: only 3 % of comments related to historical & cultural value, while 60 % reflected recreational value. This dominance of recreational perceptions reflects the zoo’s primary function as a venue for animal display, education, and family leisure, rather than as a heritage site.

Overall, aesthetic and recreational values emerged as the most prominent across the 24 parks, whereas educational value consistently showed the lowest frequencies. Educational and spiritual values were relatively stable across sites, with mean proportions of 2 % and 16 %, respectively.

3.2. Correlations between different CES perception categories

Spearman correlation analysis was conducted to explore interrelationships among CES categories (Fig. 5). The strongest correlations were observed between: (1) Social interaction and spiritual value

($\rho = 0.723$, $p < 0.01$), (2) Recreational and spiritual value ($\rho = 0.666$, $p < 0.01$), and Recreational and social interaction value ($\rho = 0.623$, $p < 0.01$).

These patterns suggest that recreation, social interaction, and spiritual experiences are co-experienced rather than isolated phenomena. Visitors engaging in recreational activities such as boating, festivals, or walking frequently report feelings of sociability and belonging, which in turn strengthen spiritual well-being. This aligns with socio-ecological restoration theory, which posits that collective leisure behaviour reinforces both individual recovery and community identity. Urban parks therefore function not only as ecological assets but also as social infrastructures of shared meaning and cohesion.

Existing research supports these mechanisms. Svendsen et al. (2016) observed that park users and parks co-produce psycho-social-spiritual benefits, where activities in nature foster connection with the self, others, and the environment. Jennings and Bamkole (2019) further noted that urban green spaces are associated with positive health behaviours, including physical activity and social interaction. Similarly, Xiong et al. (2024) demonstrated that the mental-health benefits of park environments are often mediated through social interactions, underscoring the interdependence of recreational, social, and spiritual values.

Notably, aesthetic value showed weaker correlations with other categories, indicating that landscape appreciation is experienced more independently. Educational value, while consistently low in frequency, was moderately related to historical-cultural value, suggesting that interpretive signage and cultural heritage settings may act as triggers for learning. These findings highlight the multi-dimensional and overlapping nature of CES, with practical implications for integrated park design.

3.3. Sentiment analysis

3.3.1. Characteristics of different types of CES perceptions

The overall order of perceived frequencies was: recreational value > aesthetic value > spiritual value > historical & cultural value > social interaction value > educational value (Fig. 6a). Whether CES in urban parks can be perceived generally rests on whether they can be seen or used by users. Recreational value is frequently expressed through facilities for leisure, entertainment, and festivals, while aesthetic value is reflected in both landscape quality and user behaviour. The inherent beauty of natural scenery is often accentuated by thoughtful design—for example, the tulip displays in Zhongshan Park are particularly admired. Consequently, recreational and aesthetic values are the dimensions most easily recognised by users.

Sentiment tendencies were overwhelmingly positive across all

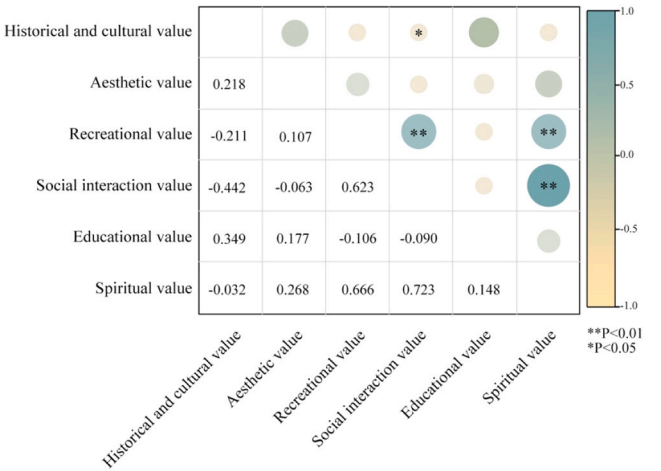


Fig. 5. Spearman correlation between categories of CES. Note: **, $P < 0.01$; *, $P < 0.05$.

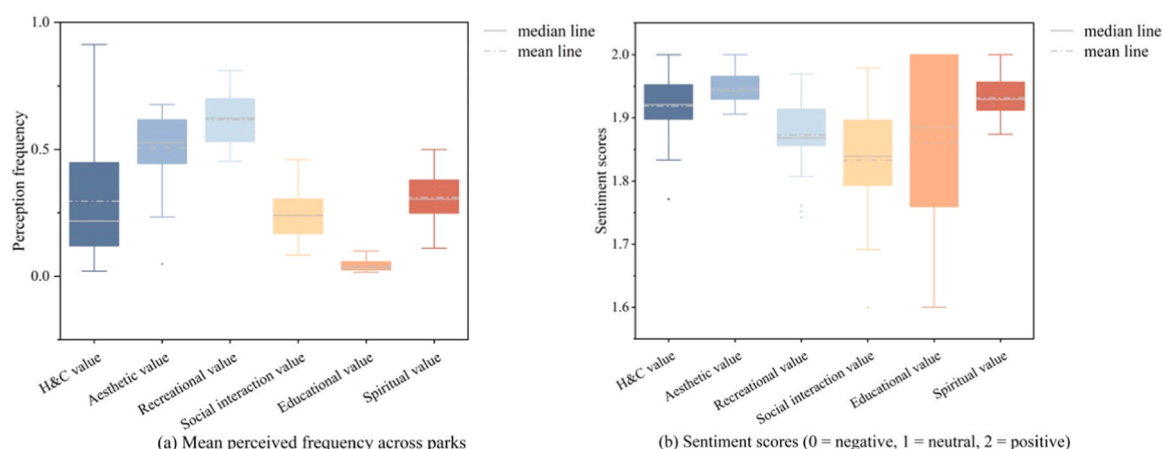


Fig. 6. Perceived frequency and sentiment tendency of CES categories. (a) Mean perceived frequency across parks; (b) Sentiment scores (0 = negative, 1 = neutral, 2 = positive).

categories (mean > 1.8 on a 0–2 scale), indicating broad satisfaction with Beijing's parks (Fig. 6b). Among them, aesthetic value received the highest ratings, underscoring the importance of ecological and visual quality to user experience. In contrast, recreational value, despite being the most frequently perceived, exhibited lower emotional appeal, suggesting that while recreational facilities are widely used, the experiential quality of recreation could be further enhanced.

Most social media comments displayed neutral or mildly positive tones, reflecting everyday satisfaction rather than extreme affective expression. In this study, neutral sentiment is not interpreted as dissatisfaction, but as an indication of stable experiences with potential for enhancement. When integrated into the IPA framework, such neutral evaluations contribute to identifying CES dimensions where experiential quality may lag behind public expectations, thereby highlighting opportunities for strategic improvement rather than problem areas.

3.3.2. Differences across park types

Differences were also evident between park types. The Kruskal–Wallis test confirmed significant variation in sentiment tendencies ($p = 0.047$, $p < 0.05$), thus rendering the initial hypothesis H0 invalid and demonstrating a substantial variation in the emotions associated with various types of parks. The Mann–Witney U test showed statistically significant differences between historical and comprehensive parks ($p = 0.011$, $p < 0.05$) and between historical and specialised parks ($p = 0.028$, $p < 0.05$) (Table 4).

Historical parks, while rich in cultural significance, sometimes generated neutral or critical comments on crowding and maintenance, lowering their sentiment scores. Comprehensive parks, in contrast, scored higher in recreation and aesthetics, reflecting diversified facilities. Specialised parks such as the Ming City Wall Relics Park often generated niche but very positive perceptions, particularly regarding heritage. Community parks, though smaller, elicited relatively stable emotional tendencies, highlighting their everyday role.

Overall, these findings reveal that park type significantly shapes

emotional evaluation: Historical parks are valued for heritage but would benefit from improved visitor-management and maintenance, whereas Comprehensive and Specialised parks deliver more balanced experiential quality.

3.4. Importance–performance analysis of CES

To integrate frequency (importance) and sentiment (performance), an IPA framework was constructed (Fig. 7a). The four quadrants indicate differentiated management priorities as follows:

- **Quadrant 1 - areas of strength (to be maintained):** Aesthetic value scored highly on both frequency and performance, confirming it the core competence of the park. Continuous investment in landscape quality, environmental facilities, and visual diversity is therefore essential.
- **Quadrant 2 - areas of hidden strengths (may be overlooked):** H&C and spiritual values were less frequently perceived but highly positive when recognised. These latent advantages can be activated through immersive interpretation, contextual restoration of heritage features, and curated cultural or contemplative events.
- **Quadrant 3 - areas of low priority (to be observed or selectively improved):** Educational and social interaction values were under-recognised and scored lower in sentiment. While not urgent, they can be strengthened by linking with other values—for example, integrating interpretive learning into heritage walks or designing multi-functional rest areas to encourage casual interaction.
- **Quadrant 4 - areas of recommended improvement (relative performance gaps to be prioritised):** Recreational value was the most frequently recognised yet only moderately satisfying dimension. This quadrant therefore signals opportunities for experiential enhancement rather than deficiencies. Upgrading comfort, accessibility, and service quality (such as through spatial audits, inclusive

Table 4
Statistical differences in sentiment tendencies across park types (Mann–Witney U test).

Mann–Witney U test ^a						
	Historical park- Comprehensive park	Historical park- Specialised park	Historical park- Community park	Comprehensive park- Specialised park	Comprehensive park- Community park	Specialised park- Community park
Mann–Witney U test	1171.000	316.500	289.000	533.500	319.000	95.000
Wilcoxon W	2656.000	1801.500	1774.000	2363.500	397.000	173.000
Z	-2.549	-2.204	-0.582	-0.077	-0.621	-0.551
Asymp. Sig. (two-tailed)	0.011	0.028	0.561	0.938	0.535	0.582

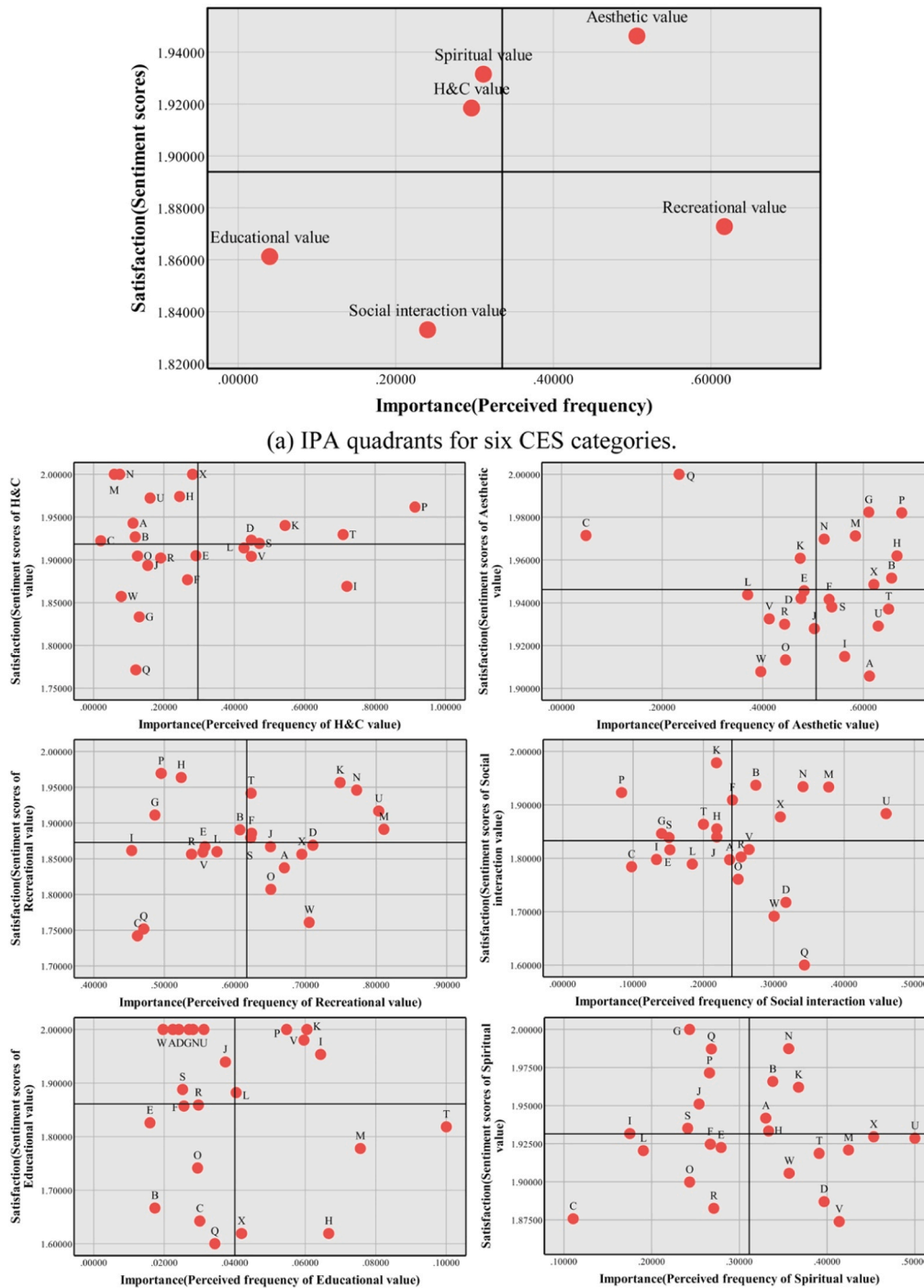


Fig. 7. Importance–Performance Analysis (IPA) of CES values and park-level results. (a) IPA quadrants for six CES categories; (b) Distribution of 24 parks by quadrant.

amenities, and responsive management) can convert routine satisfaction into higher engagement and well-being.

Fig. 7b shows park-level IPA results. Rendinghu Park, Longtan Zhonghu Park, and Yongdingmen Park appeared most often in the first quadrant, generating high levels of both attention and positive

sentiment. In contrast, Longtan Park, Beijing Zoo, and Beihai Park were positioned more frequently in the third quadrant, suggesting the need to increase the recognition of specific CES dimensions to strengthen user experience.

These findings provide clear evidence for targeted park renewal. Aesthetic quality should be maintained and diversified; historical and

spiritual values require stronger activation; education and social interaction need innovative integration; and recreation demands systematic quality upgrades. Together, these strategies support Beijing’s Garden City initiative by aligning renewal priorities with public perceptions and expectations. Table 5 synthesises management recommendations by quadrant.

4. Discussion and implications

This study contributes to a growing body of work demonstrating how big data and artificial intelligence can be used to evaluate CES in urban settings.

4.1. Identifying CES perceptions through multi-source social media data

With the rise of big data and mobile technologies, people increasingly share their everyday experiences of urban places online (Huang et al., 2022), including parks. Social media data reveal how users assign value to places (Roberts et al., 2017) and complement traditional

Table 5
Optimisation and improvement recommendations based on Importance–Performance Analysis (IPA).

Quadrant type	CES type	Recommendations
Quadrant 1: Areas of Strength	Aesthetic value	(1) Maintain consistent investment in landscape quality and environmental facilities. (2) Introduce seasonal and thematic landscape installations to enhance variety. (3) Optimise visual experiences through nighttime illumination design, creating more distinctive nocturnal landscapes.
Quadrant 2: Areas of Hidden Strength	H&C value & Spiritual value	(1) Revitalise cultural heritage through contextual restoration of pavilions, stone tablets, and garden relics, coupled with interpretive signage explaining their historical significance. (2) Curate thematic events (e.g., guided heritage walks, craft fairs, or cultural-festival programmes) to activate latent historical and spiritual meanings. (3) Cultivate spiritually restorative spaces by designing contemplative zones that foster emotional resonance and belonging (e.g., meditation courts, quiet gardens).
Quadrant 3: Areas of Low Priority	Educational value & Social interaction value	(1) Integrate educational and social functions with other CES dimensions to enrich experiences. (2) Optimise existing spaces to improve comfort and accessibility. (3) Facilitate community-led activities to strengthen informal learning and social cohesion.
Quadrant 4: Areas of Recommended Improvement	Recreational value	(1) Conduct spatial audits to identify under-used areas and redesign them for multi-use recreation (e.g., temporary markets, pop-up play zones). (2) Upgrade service infrastructure through precision-management protocols. (3) Install demand-responsive amenities, such as inclusive play and fitness equipment, shade structures, and family-friendly seating clusters, informed by real-time user-feedback tools.

surveys (Grzyb et al., 2021; Huang et al., 2022; Pickering et al., 2018), which can be costly, time-consuming, and limited in scale or demographic reach (Di Minin et al., 2015; Sessions et al., 2016; Tenkanen et al., 2017). Compared with traditional questionnaires, social media data offer wider spatial coverage, richer temporal dynamics, and timelier reflections of emotion and experience.

Previous studies have successfully used Flickr and Wikiloc photos to assess CES (Richards and Friess, 2015) and the attractiveness of parks (Norman and Pickering, 2019). Twitter data to explore emotions inside and outside parks (Plunz et al., 2019) and to investigate seasonal changes in physical activity in urban green spaces (Roberts et al., 2017) or park popularity (Tenkanen et al., 2017). Whilst previous research has predominantly utilised single social media platforms, such data can be unrepresentative due to platform-specific user demographics and varying expressive conventions. Consequently, to capture a more accurate and diverse range of public perception, this study employs a multi-platform approach, integrating data from Weibo and Xiaohongshu. This strategy is supported by established work, including Ren et al. (2025) and Liang et al. (2022), who similarly leveraged multiple sources, thereby validating the methodology.

This approach delivers three practical advantages. First, it enables faster and lower-cost evaluation of public perceptions, reducing dependence on field surveys while retaining interpretive depth. Second, by integrating data from multiple social media platforms, it mitigates the biases inherent in any single source and captures a broader spectrum of demographic and behavioural diversity. Third, it captures spontaneous, real-world expressions of value rather than post-hoc reflections, offering a more authentic representation of people’s experiences. Forth, by aligning digital trace data with planning indicators, it supports replicable, near-real-time monitoring of CES to inform ongoing park management.

4.2. Text categorisation using large language models

Previous research often relied on machine learning, such as fine-tuned RoBERTa models for assessing park accessibility and attractiveness (Zhao et al., 2025). However, these methods depend on large labelled datasets, increasing researcher workload. LLMs offer a step change: they can categorise text with few-shot or even zero-shot learning, reducing reliance on manual annotation. Studies confirm their effectiveness: Zhang et al. (2023) demonstrated the superiority of large models compared to small models in text sentiment analysis, indicating that training large models using few-shots learning alone can exceed the performance of conventional models.

Furthermore, LLMs exhibit strong generalisation capabilities and robust contextual learning skills (Rasheed et al., 2025; Yang et al., 2024). Unlike traditional machine learning, which requires task-specific data collection and annotation, LLMs acquire transferable linguistic knowledge through broad pre-training, allowing the same model to be applied across diverse text classification tasks without repeated data labelling. Additionally, their ability to process lengthy inputs and perform multi-label classification (Dagdelen et al., 2024; Sakai and Lam, 2025) makes them particularly suited to analysing complex social media reviews for CES. In this study, the selected LLM (Moonshot) demonstrated specific advantages in Chinese multi-label text classification, improving both the accuracy and efficiency of CES categorisation compared to conventional approaches.

Recent research has begun to explore LLMs in landscape and urban-greening studies—for example, evaluating park satisfaction across countries (Lai and Deal, 2025), constructing plant-species datasets for green-roof design (Han et al., 2025), and assessing street-environment quality (Cai et al., 2025). Yet these applications remain largely confined to sentiment detection or visual analysis. By contrast, the present study extends LLMs to the systematic classification of CES, linking semantic analysis directly with IPA to generate planning-relevant evidence. This integration demonstrates how AI tools

can move beyond descriptive analytics to support targeted management and policy decisions.

Overall, the LLM-based framework tested in this study delivers measurable gains in accuracy, interpretability, and scalability over earlier text-analysis methods. It offers a replicable model for large-scale, multilingual CES assessment, adaptable to other cities and digital datasets, thereby advancing the methodological frontier of urban-greening research.

4.3. CES evaluations and cross-context comparisons

Urban parks are critical ecological spaces strongly associated with human well-being (Ayala-Azcárraga et al., 2019; Wang et al., 2024; Zhou et al., 2025). Understanding and quantifying their CES is essential for garden city and eco-city development. Our analysis confirmed that aesthetic and recreational values were the most frequently perceived—similar to other survey-based studies (Bi et al., 2024; Buchel and Frantzeskaki, 2015; Ko and Son, 2018; Mao et al., 2023; Riechers et al., 2017). Conversely, educational value is the most difficult to perceive. Some parks focus more on the recreational value of environmental education resources and place less emphasis on the spatial representation of environmental education (Chi et al., 2024).

We also found strong correlations between recreational, social, and spiritual values. Visitors often experience enjoyment, sociability, and reflection simultaneously during park use. This aligns with Kwon et al. (2021), who showed that social support mediates the link between green spaces and well-being, and with Peschardt and Stigsdottir (2013), who found that “social” and “serene” sensory dimensions enhance restorativeness. Place attachment further mediates these experiences, while leisure and physical activities contribute to mental health and resilience (Castañeda et al., 2024; Nath et al., 2018; Nawrath et al., 2022). Such co-occurrence of values is particularly evident among younger users, who seek spaces that integrate active recreation, social interaction, and contact with nature (Ziaesaeidi, 2025).

Beyond Beijing, comparable CES patterns have been observed across other megacities. For instance, Wan et al. (2021), analysing Instagram posts from Hong Kong’s urban parks, reported that aesthetic appreciation, happiness, and restoration were the most salient value expressions. Similarly, Koh et al. (2022) found that Bishan-Ang Mo Kio Park in Singapore delivers abundant cultural ecosystem services, with recreation and aesthetic enjoyment as dominant experiential benefits. These findings suggest that visual pleasure and leisure fulfilment represent universal components of urban park experience in compact, high-density cities.

While Beijing’s cultural and institutional setting differs from other Chinese megacities such as Shanghai and Guangzhou, the patterns identified here (particularly the integration of heritage conservation with everyday recreation) offer transferable lessons for urban renewal and park governance within China. At the same time, Beijing’s distinctive institutional and spatial context (e.g., its governance structure, hukou-based population management, and coexistence of imperial heritage with modern redevelopment) shapes park usage in ways that differ from Hong Kong or Singapore. Yet these characteristics also reflect broader challenges faced by rapidly urbanising Asian megacities: balancing heritage conservation with densification, and ensuring equitable access to high-quality green spaces.

Accordingly, the integrated framework developed in this study, which combines LLM-based text analytics with Importance-Performance Analysis, offers a transferable methodology for assessing public perceptions of CES across cities with varied socio-spatial systems. While the empirical findings are context-specific to Beijing, the approach contributes to a wider international dialogue on how cultural ecosystem services underpin urban greening, community well-being, and sustainable city development.

4.4. Implications for renewing and managing urban parks in Beijing

Incorporating public perceptions into park renewal is increasingly recognised as a way to support well-being and social equity (Dade et al., 2020). Zheng et al. (2023) focused on Cultural Ecosystem Services in the ecological space of Beijing’s core districts, and proposed relevant spatial management policy recommendations and ecological spatial planning suggestions. Therefore, this study explores the perception of CES in different types of parks to provide a valuable reference for future urban park renewal and operation. We found that different types of parks have their own preferences for CES perceptions, mainly influenced by the characteristics of the parks themselves. Based on the characteristics of the park and CES preferences, park administrators should develop appropriate strategies to enhance CES with lower perceived intensity to strengthen the comprehensive benefits of the park.

Our IPA highlights where improvements are most needed. Aesthetic value emerged as a major strength, already highly perceived and positively evaluated, indicating that continuous investment in landscape design and maintenance is essential. Hidden strengths such as historical-cultural and spiritual values could be better activated through immersive interpretation and contemplative spaces. Educational and social interaction values, currently under-recognised, should be integrated with other functions—for example, by linking cultural heritage interpretation with environmental education or enhancing facilities that support informal social encounters. Recreational value, although widely recognised, requires significant improvement in quality, comfort, and inclusiveness, making it the top priority for renewal.

In conjunction with the IPA, for historical & cultural value, the perceived frequency of several significant historical parks, including Zhongshan Park, Taoranting Park, and Temple of Moon Park, are situated in the fourth quadrant. This suggests that user satisfaction requires enhancement, necessitating a focus on strategies to improve visitor contentment in the management and modernization of these parks. For aesthetic value, most of the parks are perceived by visitors and having a high level of satisfaction, with Nanguan Park and Shichahai Lake Park need to focus on improving visitor satisfaction. To enhance the value of recreation and aesthetic, Beijing Zoo, and Jingshan Park should incorporate recreational facilities or programs that align with the historical and cultural significance, thereby augmenting users’ appreciation of these attributes. For social interaction value, larger city parks focused on history and culture need to be optimized and upgraded, such as Beijing Zoo, Zhongshan Park, Beihai Park, and Temple of Heaven Park. Ultimately, regarding educational and spiritual qualities, most parks, except Beijing Zoo and Dongdan Park, exhibited strong positive affective tendencies; hence, these two parks should consider measures to enhance satisfaction in these areas moving forward.

Table 5 presents a synthesis of the IPA findings and renewal strategies. This evidence-driven mapping offers actionable guidance for park managers and planners, ensuring that limited resources are allocated to dimensions most in need of improvement (Xu et al., 2025; Yang and Zhang, 2025). For instance, investments in play and sports infrastructure can strengthen recreational value, while integrating interpretive signage, outdoor classrooms, or nature-based learning programmes can enhance educational functions. Similarly, promoting community events or participatory activities within parks can increase opportunities for social interaction.

Importantly, these recommendations align with the goals of Beijing’s Garden City initiative, which aspires to create equitable, inclusive, and resilient urban green spaces. By identifying which CES dimensions require targeted interventions, this study provides a replicable framework for park renewal that goes beyond aesthetic greening to emphasise human-centred values. Such a framework can help ensure that the Garden City initiative supports not only ecological performance but also the well-being and cultural identity of urban residents.

4.5. Limitations and prospects

While this study demonstrates the value of integrating social media analytics with LLMs for assessing CES, several limitations should be acknowledged. First, the analysis is subject to platform and demographic biases (see, e.g., Ling et al., 2024; Yang, 2020). User-generated content from Weibo and Xiaohongshu does not fully represent the broader population, as each platform attracts specific user groups and algorithmic recommendation mechanisms may amplify visually engaging content. As a result, aesthetic and recreational experiences may be over-represented, while educational or heritage-related values receive less visibility, particularly for smaller community parks. Future research could mitigate these biases through triangulation with surveys, participatory mapping, or additional data sources.

Second, the IPA-based interpretation focuses on aggregated patterns of perceived importance and sentiment rather than isolated negative experiences. Although explicitly negative comments can offer valuable insights into site-specific operational issues, they account for a relatively small and episodic share of the dataset and were therefore not used to define system-level management priorities. Future studies could combine IPA with targeted qualitative analyses of negative sentiment to support more fine-grained, park-specific interventions.

Third, methodological constraints remain in applying LLM-based classification. Although validation results indicate robust performance, outputs remain sensitive to prompt design and may not fully capture nuanced meanings in short, informal texts. In addition, transferability to other languages, platforms, or governance contexts requires further testing. Finally, this study is based on cross-sectional data from 2023 to 2024 and does not capture temporal dynamics in CES perceptions, such as seasonal variation or responses to major events.

Future research could address these limitations by (1) triangulating social media findings with on-site surveys or participatory mapping, thereby reducing demographic bias; (2) extending to multimodal social media data (e.g., photos, videos) to capture more expressive dimensions of CES; and (3) conducting longitudinal studies to explore how CES perceptions shift over time in relation to social and environmental change. These extensions would not only refine methodological robustness but also provide deeper insights for capturing the evolving socio-ecological significance of urban parks. By acknowledging these limitations while outlining clear pathways for advancement, the framework proposed here offers a transparent and scalable foundation for future comparative and longitudinal CES research, contributing to the integration of human perception analytics into evidence-based urban-green-space planning.

5. Conclusion

This study quantified the cultural ecosystem services (CES) of 24 parks in Beijing's core districts through the integration of social media data and large language models. Analysing more than 46,000 user-generated comments provided a fine-grained picture of how residents perceive and emotionally evaluate different CES dimensions. Aesthetic value consistently emerged as the most salient and positively perceived dimension, while recreational, social interaction, and educational values showed lower levels of satisfaction relative to their importance, pointing to clear priorities for future park renewal.

Methodologically, the research demonstrates how LLMs can enable large-scale, timely, and reliable CES assessment with minimal manual coding. The Moonshot model proved particularly effective for Chinese text classification, achieving high accuracy while substantially reducing labour requirements. This approach enriches the methodological toolkit for CES evaluation and creates capacity for continuous, real-time monitoring of public perceptions as they evolve alongside urban change.

Substantively, the findings underline that CES are not experienced in isolation but as interlinked domains: recreation, social interaction, and spiritual values frequently co-occur, shaping experiences of parks as

social and restorative spaces. Recognising these interdependencies can help planners move beyond single-value assessments and design park environments that may foster multiple, overlapping dimensions of well-being.

For Beijing, the results provide concrete evidence to guide the Garden City initiative. Improving recreational quality, expanding opportunities for education, and creating inclusive social spaces should be prioritised, while maintaining aesthetic and ecological excellence as enduring strengths. More broadly, the integrated framework presented here offers a transferable model for assessing public perceptions of urban green space in other high-density cities. By combining empirical rigour with actionable insights, this study reinforces the case for embedding cultural ecosystem services within mainstream urban planning and demonstrates the value of data-driven approaches for capturing the lived experiences that make parks indispensable to sustainable urban life.

CRedit authorship contribution statement

Shanwen Zheng: Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Yuyue Ren:** Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Chaoyang Zhu:** Validation, Supervision, Methodology, Formal analysis. **Xuying Wang:** Writing – review & editing. **Mingmin Zhao:** Writing – review & editing. **Shihan Wang:** Writing – review & editing. **Jianqiang Wang:** Supervision, Resources, Project administration. **Tianren Yang:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ufug.2026.129263](https://doi.org/10.1016/j.ufug.2026.129263).

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